

ZEPING ZHANG

Ph.D. Candidate and AI Scientist at the University of Ottawa specializing in AI: World Model, LM and autonomous driving. Proven track record in developing SOTA future forecast models and lightweight sensor fusion networks in a fast-paced environment. Expert in C++ (ACM-ICPC competitive level) and PyTorch. Passionate about pushing the frontiers of AI through physics-aware deep learning and large-scale simulation.

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EDUCATION

2023-Present Ph.D. Computer Science(in AI) | University of Ottawa | Canada - Estimated Graduate Time: May 2027

2021-2022 M.A.sc ECE with Concentration in AI | University of Ottawa | Canada

2017-2021 B.sc Computer Science | Nanjing University | China

RESEARCH BACKGROUND

University of Ottawa | Ph.D. Researcher

September 2023 – Present

Project: Multi-Source 3D Data Fusion for Railroad Monitoring (NRC Collaboration)

- Developed an ego-decoupled generative model using conditional flow matching to efficiently predict future 3D environments, overcoming latency bottlenecks for autonomous driving perception.

Project: Autonomous Lunar Rover Occupancy Detection (MDA Collaboration)

- Designed **Lunar-ODNet**, a lightweight 3D occupancy network (0.22M parameters) achieving **0.841 mIoU** on simulated lunar terrains.

Project: Physics-Aware Diffusion for LiDAR Densification (IROS 2026 Oral)

- Proposed a fast, scanline-consistent diffusion model to densify sparse LiDAR sweeps by injecting acquisition geometry into the conditioning with ray-consistency loss, significantly outperforms existing baselines by achieving a higher coverage of **86.63%(+9.78%)** and up to **73.22% faster** than existing baselines.[2]

University of Ottawa | M.A.sc Researcher

January 2022 – April 2023

During my M.A.sc at uOttawa, our work includes 1. **an instance-level LiDAR quality booster**, which was quick accepted for publication at **RAL** and presented as an **oral in ICRA 2023**[3]. And 2. a detector-independent framework that enhances 3D object detection by utilizing an Intra-Frustum Segmentation Transformer (IFST) and a Mesh Depth Completion Network (MDCNet) to perform up to 12.2% performance gains on major benchmarks and reach SOTA results in pedestrian detection while being significantly faster. [4].

SELECTED PUBLICATIONS

- [Accepted (poster) by NIPS 2026]** Zhang, Zeping, Zhuoya Zhao , Samy Metari, Robert Laganière. *FreeOcc: Decoupling Ego-Motion for Efficient 4D Occupancy Forecasting via Continuous Flow Matching*.
- [Under Review for WACV 2027]** Zhang, Zeping, Robert Laganière. *UFMF: Radar-Guided Kinematic Priors for Real-Time 4D Occupancy Forecasting*
- [Accepted for IROS 2026]** Zhang, Zeping, Robert Laganière. *Physics-Aware Diffusion for LiDAR Point Cloud Densification*.
- [Accepted for CRV 2026]** Zhang, Zeping, Rowan Hussein, Robert Laganière. *Cross-Attention-Driven Quality Assessment for Robust Camera-LiDAR Sensor Balance*.
- Zhang, Zeping, Tianran Liu, and Robert Laganière. "Use Your Imagination: A Detector-Independent Approach for LiDAR Quality Booster." *ICRA ORAL & IEEE Robotics and Automation Letters* 8.8 (2023): 4402-4409
- Tianran Liu, Zeping Zhang, Morteza Mousa Pasandi, Robert Laganière. "What You See Is What You Detect: Towards better Object Densification in 3D detection." <https://arxiv.org/abs/2310.17842>

EXTERNAL FUNDINGS(SELECTED)

1. **Research Collaboration | National Research Council of Canada**

2025-Now

Project: FreeOcc - Real-Time 4D Occupancy Forecasting World Model.

Developed an ego-decoupled generative model using conditional flow matching to efficiently predict future 3D environments, overcoming latency bottlenecks for autonomous driving perception.

2. **Federal Research Grant for AI & IoV | Algoma University**

2025

Project: LLM-Driven Strategic Planning and Routing Optimization for Urban Autonomous Driving.

Awarded federal funding to independently investigate how Large Language Models can process IoT/IoV data to optimize vehicle routing and maximize urban traffic efficiency.

3. **AWS Activate Funding, Amazon**

2025-Now

4. Research Collaboration | NXP Semiconductors

2024

Project: Multi-Modal Pedestrian Perception: Designed and deployed a robust sensor-fusion model integrating LiDAR point clouds and RGB images for real-time pedestrian detection and distance estimation at traffic intersections.

Achieved an outstanding 97.6% detection accuracy on the internal benchmark, significantly outperforming the traditional YOLOv10 baseline (78.1%) by 19.5%. Delivered the highly accurate perception pipeline as part of a smart city initiative to enhance urban intersection safety and traffic monitoring.

5. Research Collaboration | LeddarTech

2024

Project: 3D Perception on Leddar PixSet:

Partnered with LeddarTech under an industry-sponsored initiative to deploy and evaluate advanced 3D perception algorithms on the PixSet autonomous driving dataset.

6. Research Collaboration | Synopsis

2022

Project - LiDAR Quality Booster (UYI): Developed "Use Your Imagination" (UYI), a detector-independent LiDAR quality booster.

This system uses a GAN-based cross-modality module to infer dense LiDAR shapes from image inputs. Paper accepted by ICRA 2023(Oral).

Honors & Awards

1. International Doctoral Scholarship
2. Admission Scholarship | University of Ottawa
3. First-Class Academic Scholarship (Top 5%)

2023 – Now, Canada

2023 – Now, Canada

2020, China

WORK EXPERIENCE

National Research Council – Research Collaborator

Ottawa, ON

Apr 2026 – Now

- Engineered a continuous-time 4D world model leveraging geometric ego-motion decoupling to generate accurate, physics-constrained spatial predictions for real-time autonomous navigation.

Flipboard – Data Scientist

Beijing, Greater China Feb 2020 – Jul 2021

- Improved news recommendation flows by integrating Large Language Models (LLMs) and Transformer-based modules into multi-task learning embeddings.

Thanks to my experience as an ACM contest team member during my undergraduate. I can write competitive-level C/C++ code, which is a big benefit in practical AI, with a strong mathematical background of A to A+ for most of my mathematical courses.

Technical & AI Automation Skills

- Agentic Workflows & AI Tooling: Highly proficient in orchestrating multi-agent systems and utilizing agent development kits (Cursor, Antigravity) for automated code execution, parallel task management, and complex problem-solving.
- MLOps & PM & Tracking: Weights & Biases (W&B), LangChain, Jira, Confluence. Robust experience in managing distributed model training and evaluation pipelines.
- Programming & Frameworks: C/C++ (ACM-ICPC competitive level), Python, PyTorch, TensorFlow.

English: Full Professional Proficiency; Chinese: Native

I am also a Canadian certified pilot. :)